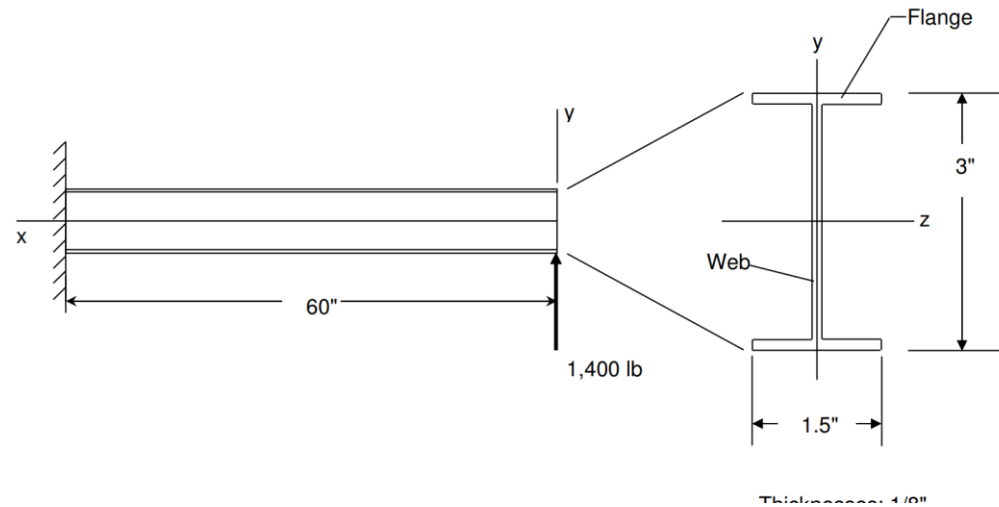
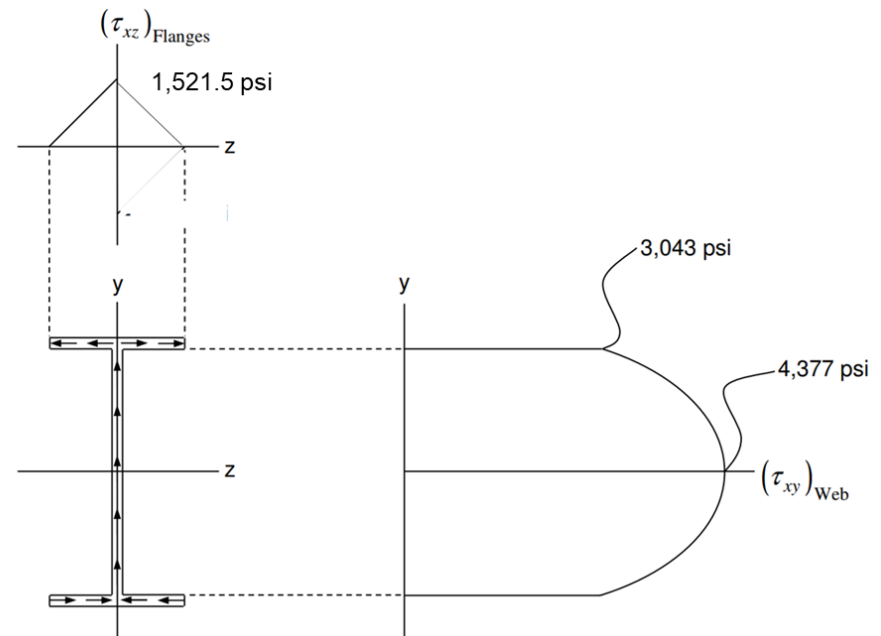


Correction to Section 4.3 Example

- Recall how Section 4.3 began with an example that looked like this:

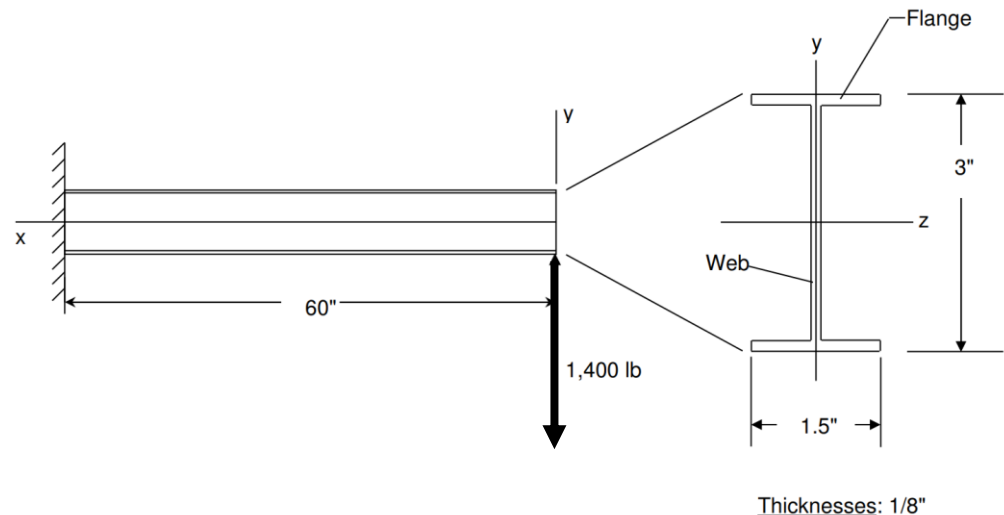


- Which led to a shear stress (or shear flow) distribution that looked like this:



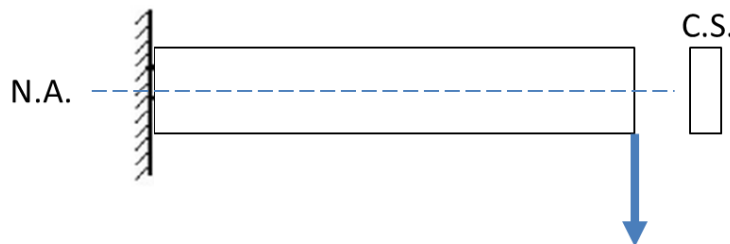
Correction to Section 4.3 Example

- However, in calculating these values, Radosta assumed internal shear force was $V = 1400\text{lb}$
- But if we were to draw a simple shear force diagram of this scenario, we would easily see that $V = -1400\text{lb}$!
- So the easiest way to rectify this is to keep all of the results, but reverse the direction of the shear force:



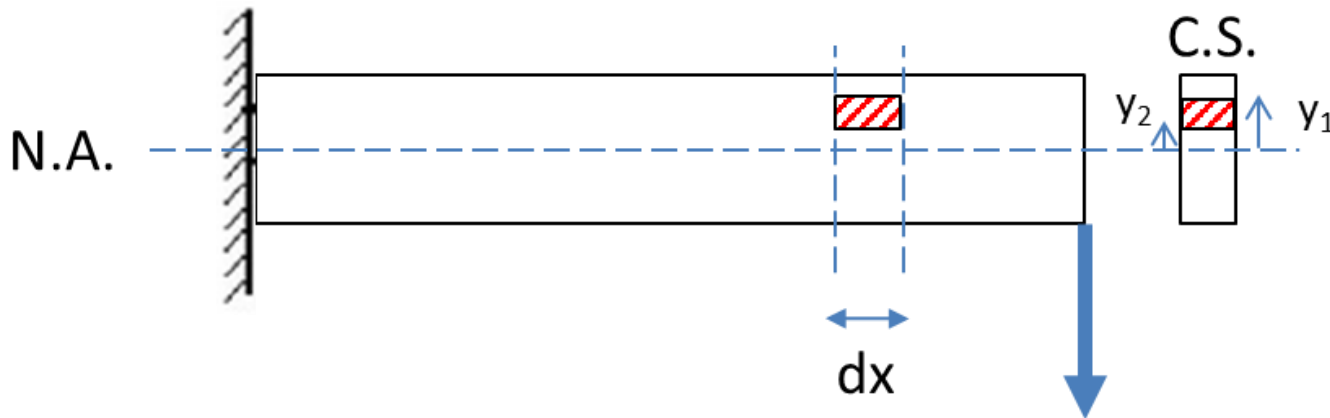
4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- So what the heck are q_{in} and q_{out} anyway?
- The familiar shear stress equation is simply $\tau = \frac{VQ}{Ib}$; there doesn't seem to be a concept of τ_{in} and τ_{out} , so where exactly did q_{in} and q_{out} come from?
- Actually there is in fact a concept of τ_{in} and τ_{out} , we just don't normally think of it that way
- Consider a beam fixed at one end with a rectangular (i.e. symmetric) cross-section that is subjected to a transverse load at the free end:

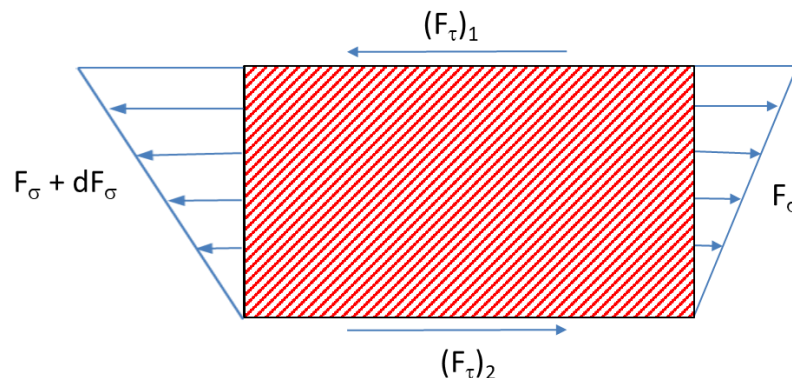


4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- Suppose we select a section of the beam with differential length dx and whose top and bottom are a distance y_1 and y_2 from the neutral axis, as shown:

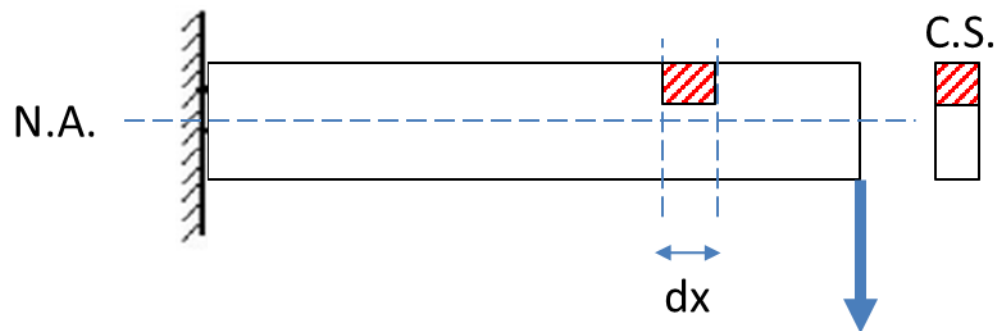


- If we zoom in on the element, we can sketch the FBD with internal forces due to normal and shear stress acting on it (F_σ and F_τ , respectively):



4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- Summing forces in the x direction and setting them = 0 yields $(F_\tau)_2 - (F_\tau)_1 = dF_\sigma$
- We'll skip a bunch of steps here, but proceeding similarly to our derivation of the GSFF last time yields $\tau_2 - \tau_1 = \frac{VQ_{12}}{Ib}$, where V , I , and b have their usual meaning, and Q_{12} is the first moment of area of the shaded part of the cross section
- Now suppose that the top of this section, instead of a distance y_1 from the neutral axis, is all the way at the top of the cross section as shown:



4.4: Shear Flow in Thin-Walled Beams Due to Transverse Loads

- In this case the shear stress at the top will be **zero** because shear stress is always zero on the free surface of a structure (i.e. it is internal only)
- Thus the equation reduces to the familiar $\tau = \frac{VQ}{Ib}$, where here Q is the first moment of area of the shaded part of the cross section shown above (stretching to the top of the cross section)
- The same is true for the GSFF above $\rightarrow$ when one end of the section under consideration is a free surface, the equation reduces to:

$$q = \frac{\left(I_y V_y - I_{yz} V_z\right) \bar{y}' A' + \left(-I_{yz} V_y + I_z V_z\right) \bar{z}' A'}{I_y I_z - I_{yz}^2}$$



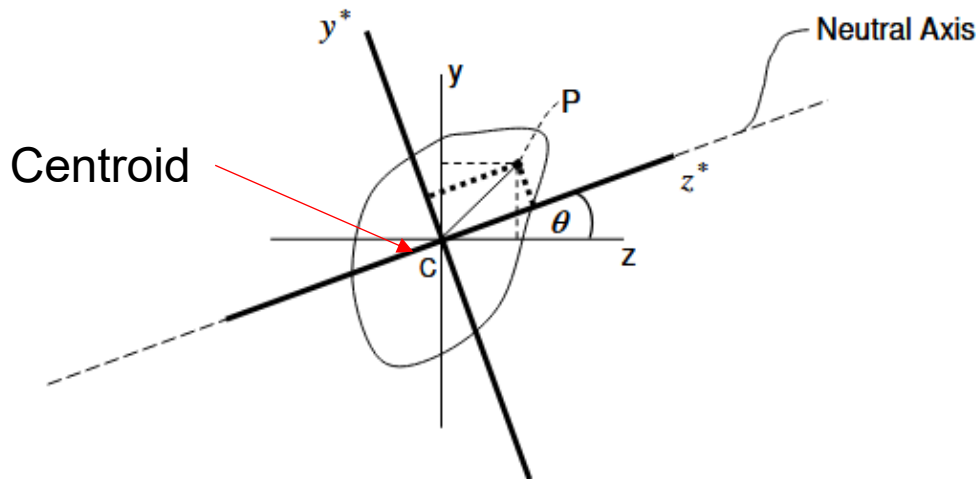
4.5: Largest Shear Flow in Beams

- We know that when $\tau = \frac{VQ}{Ib}$ is valid (i.e. $V_z = I_{yz} = 0$), maximum shear stress is at the neutral axis
- What about in the general case, i.e. when V_y , V_z , and I_{yz} are all generally nonzero?
- We saw from the General Flexure Formula that max normal stress is at the maximum distance away from the neutral axis for any general case
- Can we make an analogous proof about shear stress in the general case, as to the location of maximum shear stress?



4.5: Largest Shear Flow in Beams

- As we did before, let us again draw a general beam cross-section with standard y - z axes placed at the centroid, as well as y^* - z^* axes, rotated from y - z such that the z^* axis is along the neutral axis:



For the derivation that follows, this cross section needs to have one of its dimensions significantly thinner than the other (textbook does not mention this!)

- The General Shear Flow Formula is valid for any y - z coordinate frame with origin at the centroid; applying this to the y^* - z^* frame yields (for a section with a free surface at one end):

$$q = \frac{\left(I_{y^*}V_{y^*} - I_{y^*z^*}V_{z^*}\right)(\bar{y}'A')^* + \left(-I_{y^*z^*}V_{y^*} + I_{z^*}V_{z^*}\right)(\bar{z}'A')^*}{I_{y^*}I_{z^*} - I_{y^*z^*}^2}$$

4.5: Largest Shear Flow in Beams

- Since the z^* axis is the N.A., σ_x must = 0 when $y^* = 0$; we saw earlier that this implies

$$M_{y^*} = -\left(\frac{I_{y^*z^*}}{I_{z^*}}\right)M_{z^*}$$

- Taking the derivative of both sides of this equation with respect to x yields

$$\frac{d(M_{y^*})}{dx} = -\left(\frac{I_{y^*z^*}}{I_{z^*}}\right)\frac{d(M_{z^*})}{dx}$$

Assumes untapered cross section (i.e. I_z and I_{yz} constant with x)

- Recall above we said that $V_y = -\frac{dM_z}{dx}$ and $V_z = \frac{dM_y}{dx}$
- Inserting both of these into the above equation yields $V_{z^*} = \left(\frac{I_{y^*z^*}}{I_{z^*}}\right)V_{y^*}$



4.5: Largest Shear Flow in Beams

- Now inserting this expression for V_{y^*} back into the “ $q_{out} - q_{in}$ ” equation yields:

$$\Rightarrow q = \frac{\left[I_{y^*} V_{y^*} - I_{y^* z^*} \left(\frac{I_{y^* z^*}}{I_{z^*}} V_{y^*} \right) \right] (\bar{y}' A')^* + \left[-I_{y^* z^*} V_{y^*} + I_{z^*} \left(\frac{I_{y^* z^*}}{I_{z^*}} V_{y^*} \right) \right] (\bar{z}' A')^*}{I_{y^*} I_{z^*} - I_{y^* z^*}^2}$$

$$\Rightarrow q = \frac{V_{y^*} \left\{ \left[I_{y^*} - \frac{(I_{y^* z^*})^2}{I_{z^*}} \right] (\bar{y}' A')^* + \left[-I_{y^* z^*} + I_{y^* z^*} \right] (\bar{z}' A')^* \right\}}{I_{y^*} I_{z^*} - I_{y^* z^*}^2}$$

= 0

$$\Rightarrow q = \frac{V_{y^*} \left\{ \left[I_{y^*} I_{z^*} - (I_{y^* z^*})^2 \right] (\bar{y}' A')^* \right\}}{I_{z^*} (I_{y^*} I_{z^*} - I_{y^* z^*}^2)}$$

4.5: Largest Shear Flow in Beams

- Continuing to simplify yields:

$$\Rightarrow q = \frac{V_{y^*} (\bar{y}' A')^*}{I_{z^*}}$$

$$\Rightarrow q = \frac{V_{y^*} Q^*}{I_{z^*}}$$

- Thus we can express shear flow in this simplified form if we align the z axis with the N.A.
- Because this equation is in the same form as $q = \frac{VQ}{I}$ (i.e. $\tau = \frac{VQ}{Ib}$), we know **the largest value of q occurs along the neutral axis**

4.5: Largest Shear Flow in Beams

- **The above statement is true, in general, if the applied loads causing bending are transverse forces only**
- **If couples (i.e. pure moments with no net force) are also applied, the neutral axis may be altered**
- **In that case, the largest value of q will occur at the neutral axis that results when the couples are removed**

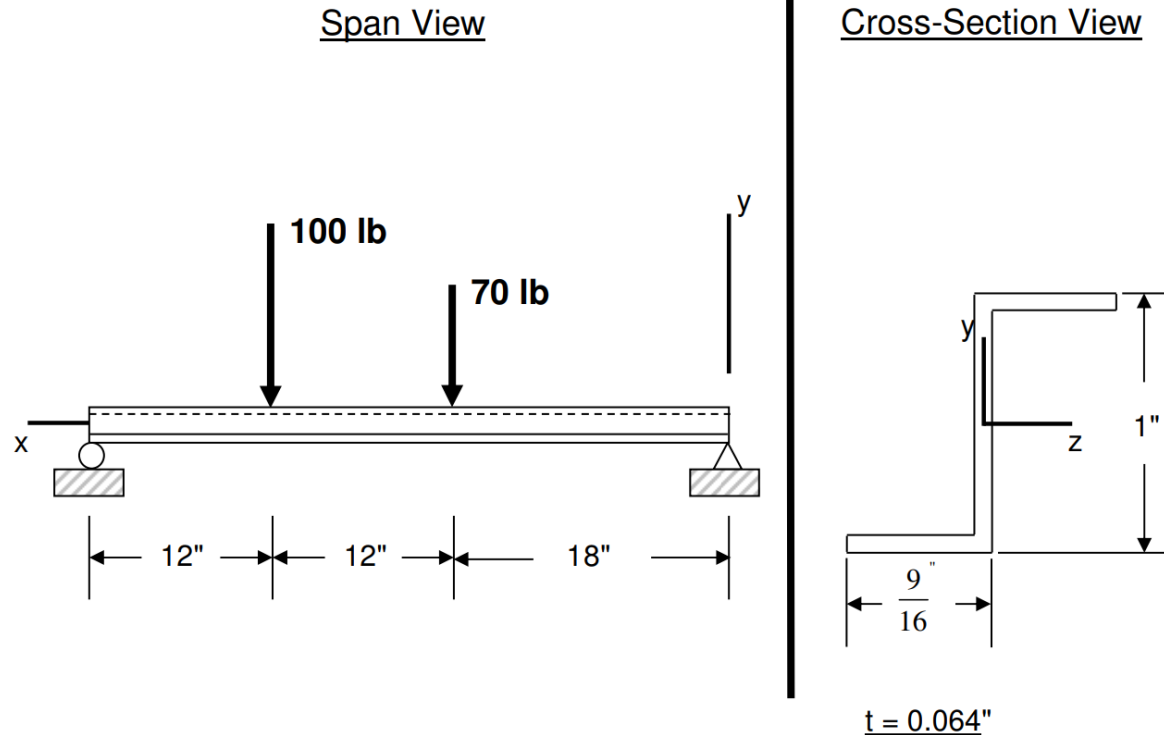


Example 4.5.1

- The simply-supported z-beam below has the same shape and dimensions as the z-beam of an earlier example

(a) Find the location and magnitude of the largest bending stress

(b) Find the location and magnitude of the largest shear flow



Example 4.5.1

- In the sketch of the beam, the x , y , z axis directions are chosen consistent with the last couple of examples (x is chosen in the axial direction positive from right to left, y and z are in the plane of the cross section as shown)
- For these chosen axis directions, neither the y nor z axis is necessarily along the neutral axis (because we don't know the neutral axis yet)
- Therefore we must use the **General Flexure Formula (GFF)** to find σ and the **General Shear Flow Formula (GSFF)** to find q , since $I_{yz} \neq 0$



Example 4.5.1

- **Here are the basic steps to solve Part (a):**
 - Choose temporary y and z axes on the cross section to solve the centroid
 - Move y and z axes to the centroid and solve I_y , I_z , and I_{yz}
 - Draw shear and bending moment diagrams to find max M along the beam
 - Use the GFF to find the neutral axis
 - Identify the critical points (i.e. points farthest from the neutral axis) and calculate σ_x at those points
- **Note that the example we worked on Mar 4 had the exact same cross section, so we already know the centroid location and the inertias with respect to that location:**

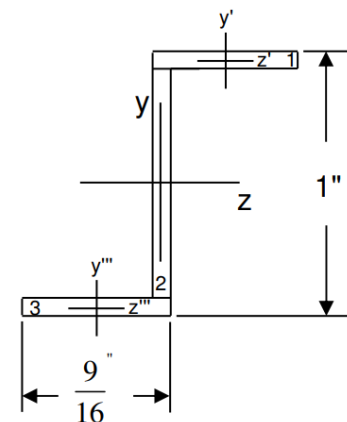
$$\bar{y} = 0.5''$$

$$\bar{z} = 0.5305''$$

$$I_z = 0.01933 \text{ in}^4$$

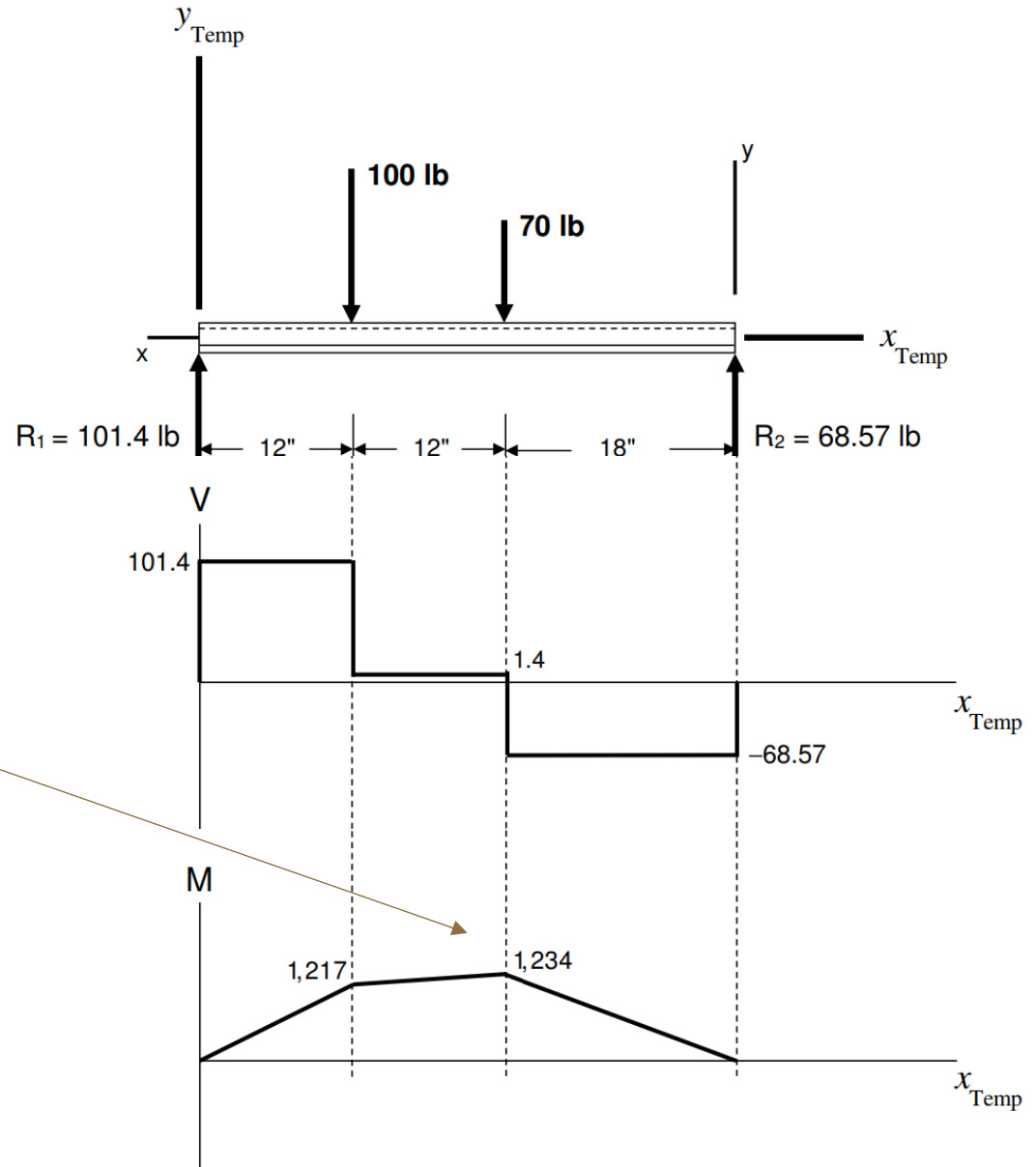
$$I_y = 0.006392 \text{ in}^4$$

$$I_{yz} = 0.008400 \text{ in}^4$$



Example 4.5.1

- For the V and M diagrams, we follow the usual steps (note the x_{temp} and y_{temp} axes)
- Max $M = 1234 \text{ in} \cdot \text{lb}$, occurring 24 in from the left end, and it is about the z axis ($M_y = 0$)



Example 4.5.1

- Since $M_y = 0$, the GFF reduces to $\sigma_x = \frac{-yI_yM_z + zI_{yz}M_z}{I_yI_z - I_{yz}^2}$
- Inserting the values for I_y , I_z , I_{yz} , and max M_z yields

$$\sigma_x = \frac{-y \left[(0.006392 \text{ in}^4)(+1,234 \text{ in} \cdot \text{lb}) \right] + z \left[(0.008400 \text{ in}^4)(+1,234 \text{ in} \cdot \text{lb}) \right]}{(0.006392 \text{ in}^4)(0.01933 \text{ in}^4) - (0.008400 \text{ in}^4)^2}$$

$$\sigma_x = -(148,800)y + (195,600)z$$

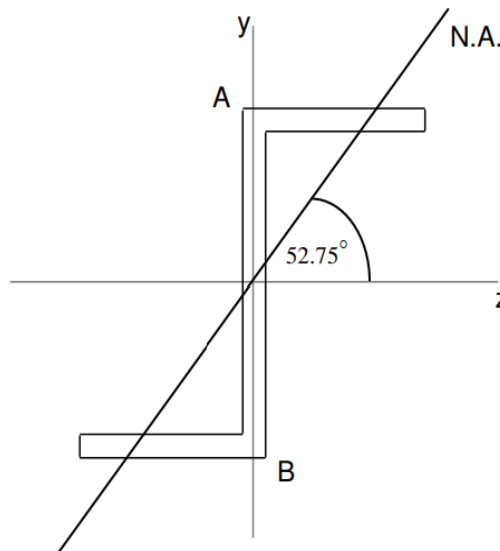


Example 4.5.1

- To find the neutral axis, set this expression = 0 to yield:

$$0 = -(148,800)y + (195,600)z \Rightarrow y = (1.315)z \Rightarrow \text{slope} = 1.315 = \tan \theta \Rightarrow \theta = 52.75^\circ$$

- Clearly points A and B are the two farthest points from the neutral axis on this cross section:



Example 4.5.1

- Inserting the y and z coordinates for each of these points into our σ_x expression yields:

Point A

(y,z) coordinates $= (0.5", -0.032")$

$$\sigma_x = -(148,800)(0.5") + (195,600)(-0.032")$$

$$(\sigma_x)_A = -80,660 \text{ psi}$$

Point B

(y,z) coordinates $= (-0.5", 0.032")$

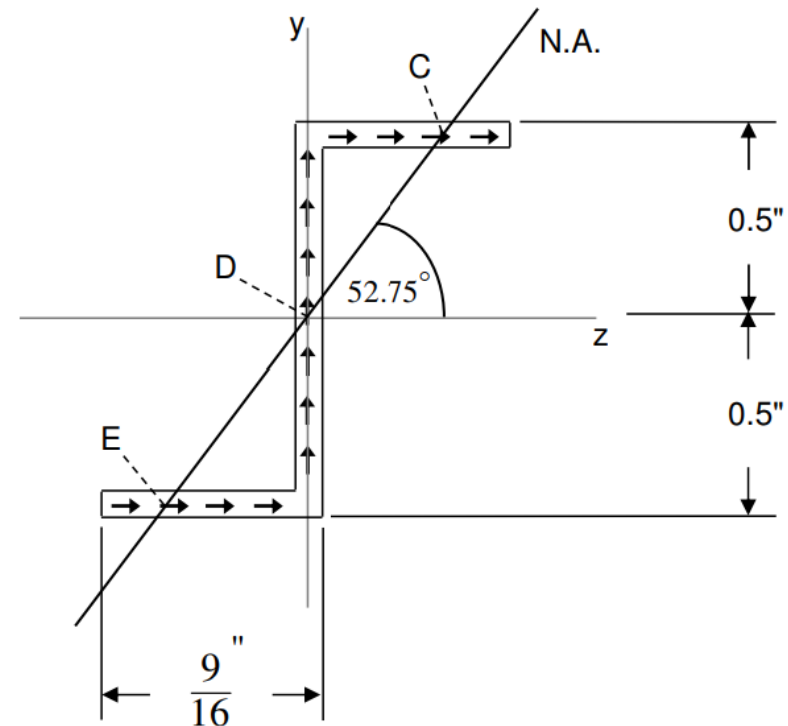
$$\sigma_x = -(148,800)(-0.5") + (195,600)(0.032")$$

$$(\sigma_x)_B = 80,660 \text{ psi}$$

- Thus, the largest bending stress is $80,660 \text{ psi}$ and occurs at both Points A and B, with Point A in compression and Point B in tension

Example 4.5.1

- Here are the basic steps to solve Part (b) in addition to what we already know from Part (a):
 - Look back at the shear force diagram to find max V along the beam
 - Identify the critical points (i.e. points on the neutral axis) and use the GSFF to calculate q at those points (will involve calculating $\bar{y}'$, $\bar{z}'$, and A')
- We know that the largest q occurs on the neutral axis; the cross-section intersects the NA at 3 points, labeled C, D, and E**
- We must find the q value at each of these 3 locations individually and choose the largest value of the 3**



Example 4.5.1

- Looking back at the shear force diagram, we see that $V_z = 0$ and $\max V$ along the beam = 101.4lb
- Let's insert these values and the values for I_y , I_z , I_{yz} into the GSFF, in its original " $q_{out} - q_{in}$ " form (we will see why later):

$$q_{out} - q_{in} = \frac{\left(I_y V_y - I_{yz} \frac{0}{z} \right) \bar{y}' A' + \left(-I_{yz} V_y + I_z \frac{0}{z} \right) \bar{z}' A'}{I_y I_z - I_{yz}^2}$$

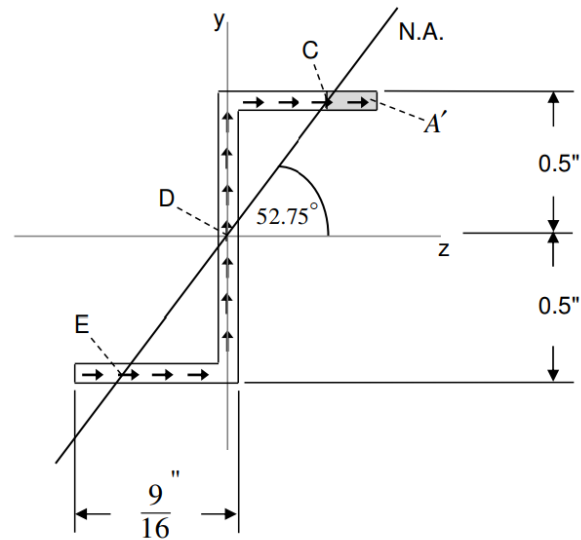
$$\Rightarrow q_{out} - q_{in} = \frac{\left\{ \begin{aligned} & \left[(0.006392 \text{ in}^4)(+101.4 \text{ lb}) \right] \bar{y}' A' \\ & + \left[-(0.008400 \text{ in}^4)(+101.4 \text{ lb}) \right] \bar{z}' A' \end{aligned} \right\}}{(0.006392 \text{ in}^4)(0.01933 \text{ in}^4) - (0.008400 \text{ in}^4)^2}$$



$$q_{out} - q_{in} = [(12,230) \bar{y}' - (16,070) \bar{z}'] A'$$

Example 4.5.1

- Let's first consider point C
- In order to calculate q_C , we need to choose a portion of the cross-section such that (a) q_C plays the role of q_{out} and (b) q_{in} is at a free surface, so that $q_{in} = 0$ (or vice versa!)
- For point C let's choose the portion shaded in gray below (i.e. all of the area to the right of point C):



Example 4.5.1

- Here we are looking at the cross-section from the right end of the beam
- Recall the sector we analyzed to derive the GSFF 
- To be consistent with this picture, we label q_{in} and q_{out} of this portion as follows 

